

Multicalibration for Unbiased Model-Based Prevalence Estimation

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Abstract

Political scientists increasingly use classifiers, including large language models (LLMs), to estimate how prevalent a concept is in a population. Standard validation reports discriminative metrics such as accuracy or the area under the ROC curve (AUC) on a labeled sample. This certifies neither unbiased prevalence estimates in the validated population nor their transport to new ones. Existing prevalence corrections, rarely used in practice, either rely on label-shift assumptions or must be re-estimated for every target. Building on Kim et al. (2022), we show that a *multicalibrated* classifier, one calibrated conditional on input features rather than only on average, yields unbiased prevalence estimates under any covariate shift that the calibrated groups can express, within their support, and needs to be fitted only once. In a simulation and two applications, an LLM coding policy topics across four countries and languages and a logistic regression predicting employment from survey data, multicalibration keeps prevalence

error below half a percentage point for shifts within the calibrated features, where standard corrections miss by around 20 points, and degrades only modestly beyond them.

Keywords: prevalence estimation, multicalibration, covariate shift, quantification, large language models, measurement validity

1 Introduction

Machine learning classifiers, and increasingly large language models (LLMs), have become standard tools for measuring political concepts from text, images and administrative records (Benoit et al. 2026; Overos et al. 2024; Mellon et al. 2024; Ornstein et al. 2025; Torres and Cantú 2022; Imai and Khanna 2016). The quantity of interest is often not the label on any single unit but an *aggregate* count or prevalence (Hopkins and King 2010; González et al. 2017; Esuli et al. 2023).

Such a classifier is typically validated by measuring how well it classifies units or agrees with human labels, using metrics such as accuracy, F_1 , AUC or intercoder reliability (e.g., Birkenmaier et al. 2024; Törnberg 2024; Knox et al. 2022; Halterman and Keith 2026; Grimmer and Stewart 2013; Barberá et al. 2021; Gilardi et al. 2023; Burnham 2025). For prevalence estimation this is not sufficient. First, a classifier can rank units almost perfectly yet misstate the proportion that are positive, because its false positives and false negatives need not cancel. The quantification literature corrects this with error rates estimated on the labeled sample (González et al. 2017). Second, corrections are tied to the population they were estimated on. When the instrument is applied to a new target, for example to compare

sub-corpora or track a quantity over time, the mix of units might change, i.e. covariates shift. If the target holds more of the cases the classifier misjudges, errors that canceled on the validation sample no longer do. That is, validity established in one context need not hold in another (Adcock and Collier 2001; Torres and Cantú 2022; Decter-Frain 2022).

In practice, most applied work takes the classifier’s labels at face value and counts them, known as Classify & Count (Esuli et al. 2023). Corrections exist, but each addresses only part of the problem, and none yields a *reusable* instrument. Adjusted variants of Classify & Count (Rogan and Gladen 1978; Saerens et al. 2002) and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023) instead carry class-conditional quantities, such as error rates or word profiles within each category, from the labeled sample to the target. That is justified under label shift, where only category frequencies change (Tasche 2023), but covariate shift generally changes those quantities (Tasche 2022). Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) require gold labels drawn from the population being analyzed, or, for the latter, a label shift or known covariate-shift weights linking the two. Importance weighting (Shimodaira 2000; Sugiyama et al. 2007; Tasche 2023) shares the covariate-shift assumption but must be re-estimated for each target, and it sets the classifier aside: its estimate is a reweighted average of hand codes, with nothing to reweight where the target sample is outside the source’s support. Related work on learned proxies (Knox et al. 2022) and on LLMs as annotators (Ziems et al. 2024) or proposed substitutes for survey respondents (Argyle et al. 2023; Bisbee et al. 2024) raises the same worry that imperfect measures distort inference; Weidmann et al. (2026) find that LLM coders deviate from experts in systematic directions.

Averaging a model’s outputs gives an unbiased prevalence estimate if the classifier is *calibrated*, that is, when its outputs or scores match empirical frequencies in the population at hand. For example, Wu and Resnick (2024) calibrate classifier outputs on a labeled sample and extrapolate to the target by assuming a stable calibration curve. But calibration is a property of one population. When the classifier’s errors vary with the features, a shift in these features that the score does not capture changes what a given score means, so the calibration curve can move (Tasche 2022). Building on Kim et al. (2022), we argue that *multicalibration*, calibration within each of a rich class of subgroups defined by the input features (Hébert-Johnson et al. 2018), can address this concern. A model, multicalibrated once on a labeled source sample, returns unbiased prevalence estimates on any target whose shift the features capture and that lies within their support.

2 Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a predictor $h(X) \in [0, 1]$. The estimand is the target prevalence $\pi^* = P^*(Y = 1)$, estimated by averaging predictions over unlabeled target observations. We assume *covariate shift*, that is, $P(X)$ may differ between source and target while $P(Y | X)$ is stable. This is in contrast to *label shift*, which holds $P(X | Y)$ fixed (Moreno-Torres et al. 2012). We also require overlap: any feature region that occurs in the target also occurs in the source. Under concept drift, where $P(Y | X)$ changes, target prevalence cannot generally be recovered without target labels.

The prevalence error of a predictor is its mean signed error, $\mathbb{E}[h(X)] - \mathbb{E}[Y] = \mathbb{E}[h(X) - Y]$,

which is not constrained by discriminative metrics. E.g. AUC depends only on how the scores rank units, so it is unchanged by any monotone transformation, yet such transformations can move the average score almost anywhere in $(0, 1)$: a classifier that ranks perfectly can misstate prevalence by any amount. Accuracy and F_1 count errors without their sign, and Classify & Count errs by the difference between its false positives and false negatives.

Calibration can correct this error. If $\mathbb{E}[Y \mid h(X) = p] = p$ for every prediction value p , then $\mathbb{E}[h(X)] = \mathbb{E}[Y]$. But calibration is a property of a specific population. Partition it into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y \mid X \in G]$ be the mean signed error in group G . Calibration ensures only that these errors cancel with the weights of the source population (i.e. $\sum_G w_G \epsilon_G = 0$). If covariate shift reweights the groups, the errors stay fixed but the weights change, and the prevalence error becomes $\mathbb{E}^*[h(X)] - \pi^* = \sum_G (w_G^* - w_G) \epsilon_G$. A classifier can therefore discriminate well and be calibrated overall while erring systematically in specific groups.

Multicalibration removes these group-level errors rather than relying on their cancellation. Let $\mathcal{G}$ contain all groups that can be formed from the calibrated features, including their interactions and intersections. A predictor f is multicalibrated with respect to $\mathcal{G}$ if $\mathbb{E}[Y \mid f(X) = v, X \in G] = v$ for every $G \in \mathcal{G}$ and prediction value v (Hébert-Johnson et al. 2018). Averaging over prediction values shows that f has zero mean signed error within every group in $\mathcal{G}$. A target shift is captured by $\mathcal{G}$ if it can be expressed as a reweighting of these groups. Because each group error is zero, reweighting the groups cannot create aggregate error: setting $\epsilon_G = 0$ in the decomposition above gives $\mathbb{E}^*[f(X)] = \pi^*$ for every target whose shift is captured by $\mathcal{G}$ and lies within the source’s support. This is the “universal adaptability”

guarantee of Kim et al. (2022), who show that such an estimate is as accurate as inverse propensity weighting, up to the multicalibration error, under any propensity model whose odds the group class can express.¹ The same predictor serves every target shift the calibrated features capture, without a new correction per target and without target labels.

This guarantee is exact only when the target-to-source density ratio lies in the span of the group indicators. Otherwise it is approximate, with error governed by how well the calibrated groups approximate that ratio; with finite calibration data, multicalibration itself also holds only approximately. Whether covariate shift and this representation condition are plausible requires substantive judgment. We obtain the multicalibrated predictor using MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

3 Simulation

We first illustrate the mechanism with a simple simulation. We use a binary feature X and a continuous signal $U \sim N(0, 1)$. The outcome model $P(Y = 1 \mid X, U) = \sigma(a_X + 1.5U)$ is the same in every population. The classifier’s score adds a logit offset for $X = 0$ only to simulate miscalibration, so it overstates this group’s probability while still discriminating well ($\text{AUC} \approx 0.89$). Each estimator is fit on 10,000 labeled samples at $P(X = 0) = 0.5$ and applied to unlabeled targets with $P(X = 0)$ from 0.01 to 0.99, over 50 repetitions. We compare multicalibration (MCGrad) to the classify and count baseline as well as the canonical Rogan-Gladen adjustment and global recalibration (Isotonic Regression). Additional details

¹Note that technically, the weaker property of *multi-accuracy*, zero mean error within each group, already suffices. We discuss this and why we recommend multicalibration in more detail in SI Section S1.8.

can be found in SI Section S4.

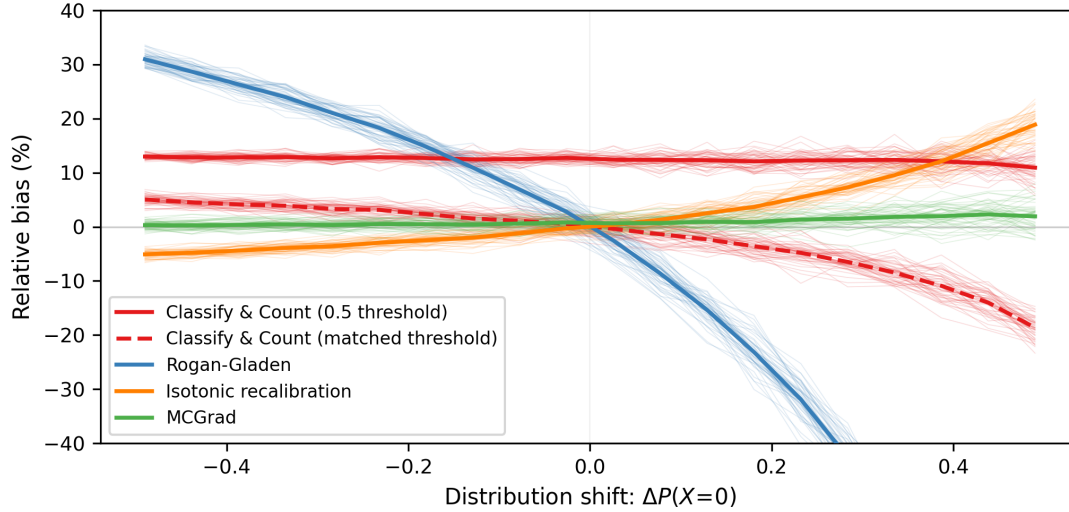


Figure 1. Relative prevalence bias under covariate shift, against the change in $P(X = 0)$ from its calibration value of 0.5. Each estimator is fit on 10,000 labeled samples. Faint lines are individual runs; thick lines are the average over 50 runs. The figure is cropped at $\pm 40\%$. The full range, all methods, and RMSE can be found in SI Sections S4–S5.

At the calibration distribution ($P(X = 0) = 0.5$), Classify & Count with the default 0.5 threshold overstates prevalence by about 12%: good discrimination does not make the counts accurate. The labeled sample can remove this bias in three common ways: by choosing the threshold so that the count reproduces the calibration prevalence, by adjusting the count for the estimated error rates (Rogan-Gladen), or by recalibrating the scores (isotonic regression). All three are unbiased at the calibration distribution, but each fits a single correction to the calibration mix of X , and the correction fails when that mix changes: the matched-threshold count drifts to about -19% , and Rogan-Gladen, whose ratio form amplifies changes in the error rates, exceeds $\pm 30\%$. Isotonic recalibration shows the mechanism most clearly: it maps a score to the same probability whatever the document’s X , which is too high for $X = 0$

documents and too low for $X = 1$. These errors cancel only at the calibration mix, so the bias grows to about +19% when nearly all documents have $X = 0$. MCGrad is fitted to the same labeled sample and the same scores, but it calibrates within each value of X , so its correction holds under shift: its average bias stays within about 2% throughout. The 0.5-threshold count, finally, is wrong everywhere; its bias barely moves only because its relative error happens to be similar in the two groups here.

4 Applications

We apply the method in two settings with known ground truth. First, Claude Opus 4.6² classifying 29,900 Comparative Agendas Project texts (Baumgartner et al. 2006) from six sub-populations in four countries and languages for whether they concern Law & Crime (CAP major topic 12), scored against the expert codes. The ground truth prevalence of the topic is 7.9% at baseline and 6–20% across scenarios. Second, a logistic regression predicting employment from 16 sociodemographic features in American Community Survey (ACS) microdata. In both, the targets’ covariate shift is known by design: within-support reweightings (of countries and document types; of age groups) and out-of-support targets (unseen document types and held-out states). The LLM returns a Yes/No label and an elicited probability (Tian et al. 2023). We multicalibrate on the yes/no label, and use the elicited probability for the methods that require a score³.

²An open-weights replication with Llama 3.3 70B (SI Section S2) reproduces the qualitative pattern and permits full reproduction.

³More details in SI Sections S2–S3

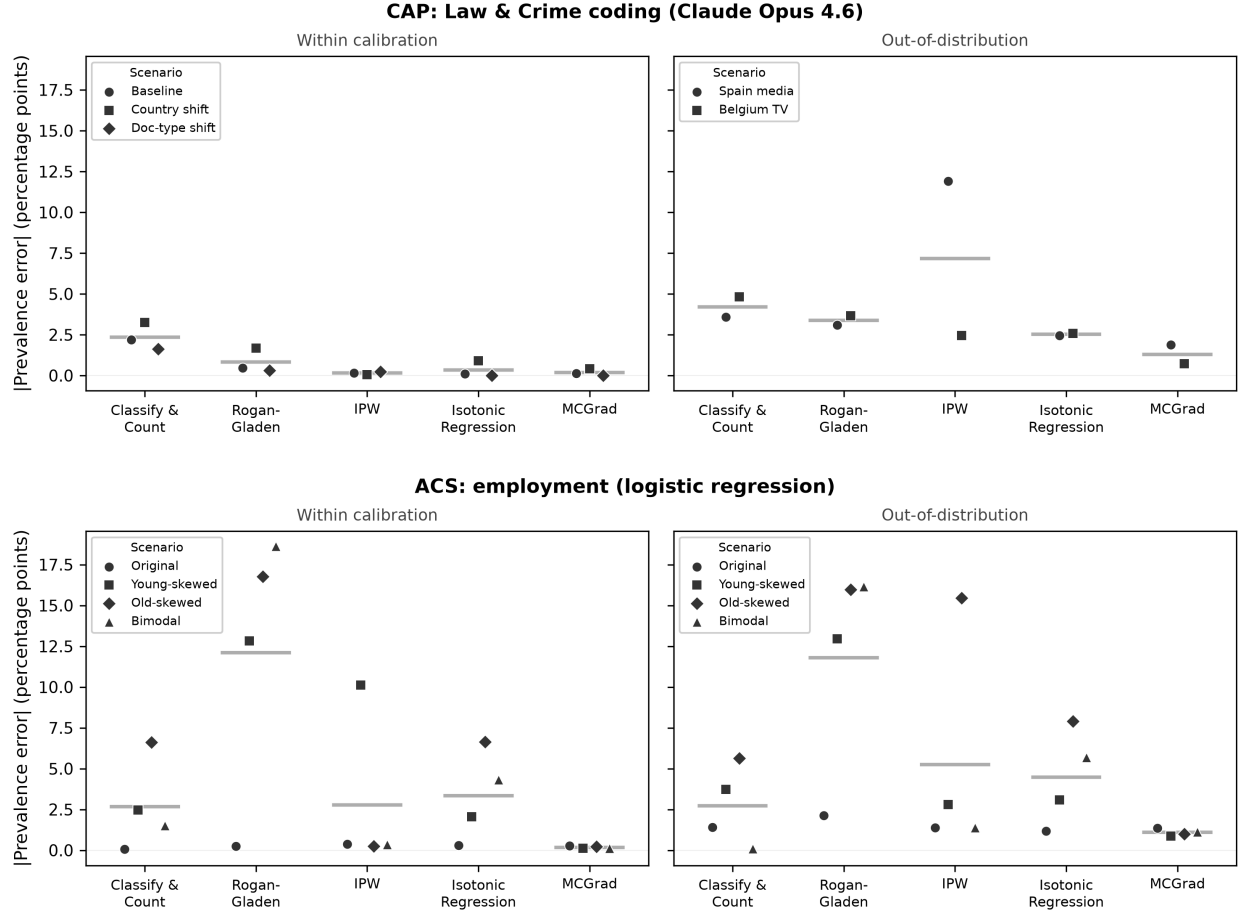


Figure 2. Absolute prevalence error (percentage points) by method and scenario. Top: CAP Law & Crime coding of 29,900 texts (1947–2018) with Claude Opus 4.6, multicalibrated on the Yes/No label. Bottom: ACS employment with logistic regression (2016–2018 microdata; eight source and six held-out states). Left: within-support targets; right: out-of-distribution targets. Markers denote scenarios and bars their means. Full numbers in SI Tables S1–S2.

The LLM shows good discriminative performance (per-language AUC about 0.98), yet Classify & Count, the applied default, errs by +1.6 to +4.8pp (Figure 2, top), overstating prevalence by nearly 30% at baseline. Rogan-Gladen reduces this within support but misses by +3.1 to +3.7pp out of it.⁴ Isotonic recalibration is accurate within support (≤ 0.9 pp) but drifts out

⁴SI Tables S1–S2 report additional methods, including SLD [sauerens2002adjusting] and PACC. ReadMe

of it (up to 2.6pp). IPW, with a gradient-boosted propensity model on the multicalibration features, is near zero within support but misses by -11.9 pp on Spanish media, whose feature values are absent from the source. Multicalibration on the bare labels stays within 0.4pp within support and 1.9pp out of it.

The survey application reproduces this ordering with a wider spread (Figure 2, bottom). Rogan-Gladen fails by 12–19pp under the age shifts, and Classify & Count and isotonic regression reach 6–8pp at the largest. IPW, using the same learner as MCGrad without tuning, is accurate on most targets but collapses on two (-10.1 and -15.5 pp): with a calibration sample thirty times larger than the target, a few source observations receive extreme weights (SI Section S3). Multicalibration stays within 0.27pp on every within-support target and 0.88–1.35pp on held-out states. Global recalibration thus suffices only when the predictor’s errors are roughly homogeneous across the groups a shift reweights, as approximately in CAP. The ACS shifts move age, which drives employment (76% employed at ages 25–54, 17% at 65+), so group errors are large and heterogeneous, and only feature-conditional calibration removes them.

5 Discussion

Discriminative validation shows only that a classifier separates classes not whether the resulting prevalence estimates are unbiased. In our LLM application, counting the labels of an LLM whose probabilities reach a per-language AUC of about 0.98 overstated prevalence

[@hopkinsking2010nonparametric], the best-known political-science quantifier, is classifier-free and fails as the label-shift diagnosis predicts, exceeding +60pp under some shifts (SI Section S7).

by 18–44% of the true rate. Multicalibration removes this bias where the classifier was validated and keeps it removed as the target shifts. Unlike importance weighting, it is fitted once, before any target is seen, and yields a single instrument for a time series or a set of sub-corpora without target labels or per-target re-estimation. It also leaves the usual workflow intact: prevalence is still the average of the classifier’s outputs, after a one-time adjustment on the labeled sample already used for validation. Label-shift corrections remain appropriate when only how common the category is changes, not what its instances look like; when the population’s composition changes along observed features, as across countries, sources or time periods, multicalibration is the reusable choice.

However, there are some limitations to this guarantee. It holds only along calibrated dimensions, so calibration data must span the anticipated variation. It also requires labeled calibration data, which in zero-shot settings reintroduces some hand-coding, though far less than per-document labeling and only once. Under mixtures the covariate-shift condition holds only approximately.

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Data Availability Statement

Replication code and data will be deposited in the Political Analysis Dataverse and are available at https://github.com/facebookresearch/multicalibrated_llm_measurement. The analyses use Comparative Agendas Project data from Denmark (Green-Pedersen and Mortensen 2019), Spain (Chaqués-Bonafont et al. 2019, n.d.), the United States (Wilkerson et al. 2025) and Belgium (Walgrave et al., n.d.), coded under the CAP master codebook (Jones et al. 2025), and American Community Survey microdata (U.S. Census Bureau 2019), accessed through the folktables package (Ding et al. 2021).

Competing Interests

All authors are employees of Meta Platforms, Inc.; Milan Vojnovic is also affiliated with the London School of Economics and Political Science. Llama 3.3, used in the replication in SI Section S2, is developed by Meta. Niek Tax, Lorenzo Perini, Fridolin Linder, Daniel Haimovich, and Milan Vojnovic are authors of MCGrad (Tax et al. 2026), the open-source

multicalibration method evaluated here. The authors declare no other competing interests.

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